

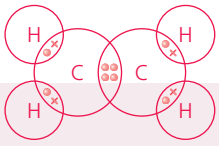
Topic 7

Fossil Fuels and Carbon Compounds

Chemistry in Context

Polymers One of the most popular toys in the world is Lego. Lego consists of colourful interlocking plastic bricks. These bricks can be assembled and connected in many ways, to construct objects such as vehicles, buildings and even working robots. Similarly, in the chemistry world, chemists join many small molecules together repeatedly to form very large molecules called polymers. Many polymers are composed of chains of carbon and hydrogen atoms alone. Ethene (C_2H_4) is one of the most important substances to make polymers. Examples of polymers include plastic toys and plastic bags.

Do you recall?



1 Draw an electron diagram of ethene, showing electrons in the outermost shells only.

2 Name ONE structural difference between an ethene and an ethane (C_2H_6) molecule.
Ethene contains carbon-carbon double bond while ethane does not.

3 Ethene for making polymers can be obtained from petroleum. Suggest ONE other use of petroleum.
As a fuel for transportation / generating electricity / heating

In this topic, you are going to study petroleum, some common carbon compounds and polymers.

Unit

25 Fossil fuels

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

Unit Key Concepts

- Origins and uses of petroleum
- Fractional distillation of petroleum
- Greenhouse effect and air pollution

25.1 Fossil fuels — a major energy source

Cooking a dinner, heating a house, lighting a street, running a factory — all these require energy. Nowadays, we are totally dependent on an abundant and uninterrupted supply of energy for living and working. According to the US Energy International Administration, global energy demand is projected to increase by 53% from 2008 to 2035 — an average annual increase of 1.6% (Fig. 25.1).

Globally, **fossil fuels** will remain the dominant source of energy. N1 As the name suggests, they are derived from the fossil remains of living organisms. The major fossil fuels are **coal**, **petroleum** and **natural gas**. N2

Petroleum is also called crude oil or oil.

N1 Refer to page T14.

N2 Do NOT use the terms 'petroleum' and 'petrol' interchangeably.

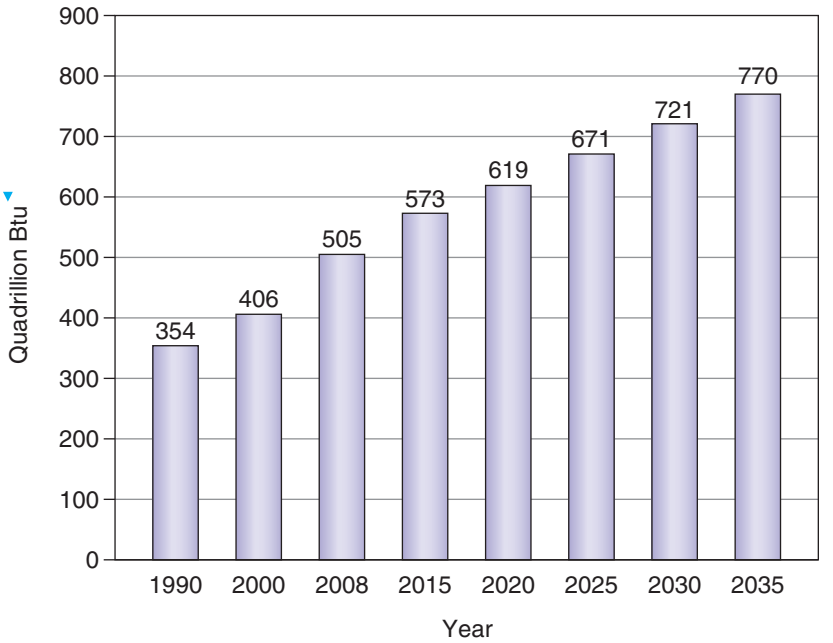


Fig. 25.1 World energy consumption (1990–2035)

Btu (the British thermal unit) is a traditional unit of energy. 1 Btu is equal to about 1 055 J.

Do you know

China's dependence on coal for energy supply

China is the world's leading consumer and producer of coal. In 2012, nearly 81% of China's electricity was produced from coal.

China plans to use less coal for electricity generation by developing other energy sources and improving the efficiency for coal consumption. Even so, the demand for coal keeps increasing as the economy grows.

fossil fuel 化石燃料　coal 煤　petroleum 石油　natural gas 天然氣

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

1 Most human activities involve using energy from fuels. List ways in which you use energy from fuels in a typical day.

2 How would our life be affected if all fossil fuels were used up?

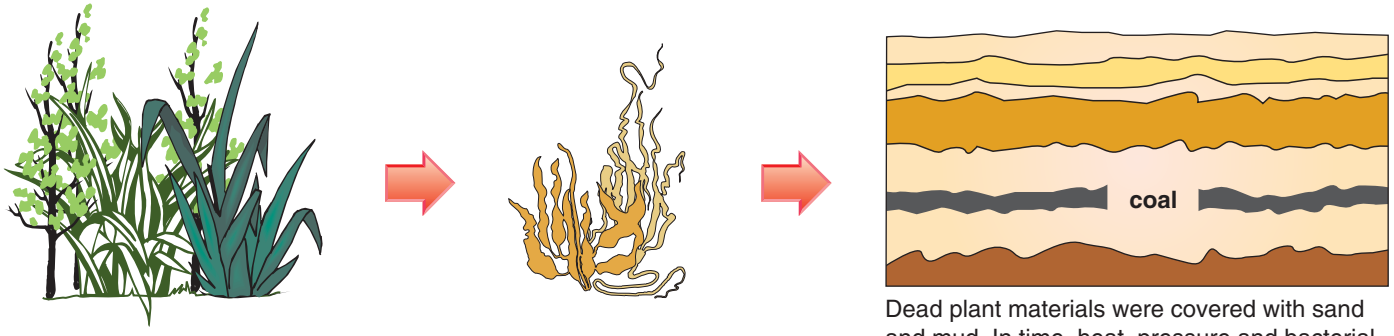
Refer to page T14.

Formation of coal



Fig. 25.2 Coal with plant fossils

Millions of years ago, *swamp forests* covered large parts of the Earth's surface. The plants died and formed thick layers of dead plant materials (Fig. 25.2). Rivers washed down sand and mud to cover the dead plant materials. In time, the combined effect of heat, pressure (from overlying layers) and bacterial action gradually changed the dead plant materials into coal (Fig. 25.3).



Plants

Plants died

Dead plant materials were covered with sand and mud. In time, heat, pressure and bacterial action gradually changed the dead plant materials into coal

Fig. 25.3 The formation of coal

Formation of petroleum and natural gas

N3 Components of natural gas
Natural gas is a mixture of methane (75–90%), ethane (5–10%), propane and butane (3–6%) and smaller amounts of other alkanes, along with a few percent of nitrogen and / or carbon dioxide and some water vapour. Natural gas is often, though not always, found together with petroleum.

Petroleum and natural gas were formed from marine animals and N3 plants that lived millions of years ago. When they died, their bodies sank to the bottom of the sea and were covered by thick layers of sand and mud.

The combined effect of heat, pressure (from overlying layers) and bacterial action gradually changed the dead remains of animals and plants into petroleum and natural gas.

swamp forest 沼澤林

N4 Refer to page T15.

Petroleum and natural gas move through *porous rocks* until they get trapped by *non-porous rocks* above them (Fig. 25.4). N4

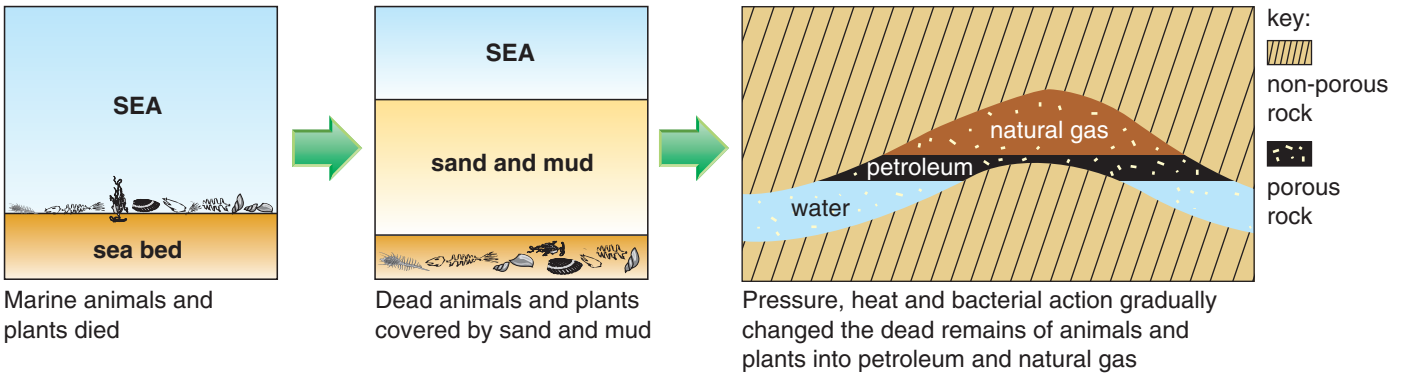


Fig. 25.4 The formation of petroleum and natural gas

Reference Website
Refer to an animation on the formation of coal, oil and natural gas:
http://resources.edb.gov.hk/~senergy/power/source_phy/flash/formation_e.html
(accessed August 2014)

25.2 Importance of petroleum

Refined petroleum has the following main uses:

- as fuels for transportation;
- as fuels for generating electricity;
- as fuels for heating; and
- as raw materials in the **petrochemical industry**.

Compounds produced from petroleum or natural gas are called **petrochemicals**. Some petrochemicals, such as detergents, *pesticides*, *pharmaceuticals* and *cosmetics* are used directly. Most petrochemicals, however, serve as raw materials in the production of other *synthetic* substances, particularly a wide range of **polymers**.

We will discuss polymers in Unit 28.

Fig. 25.5 shows the share of world petroleum consumption in 2010.

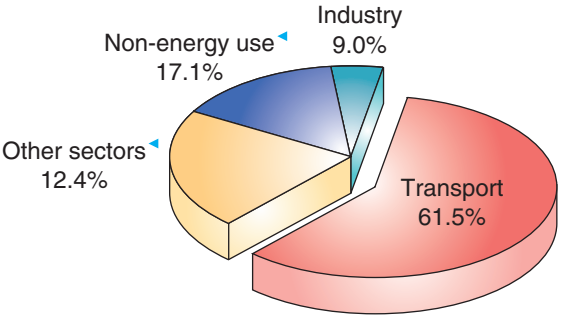


Fig. 25.5 Share of world petroleum consumption (2010)

Non-energy use covers the use of other petroleum products such as paraffin wax, lubricant, bitumen, etc.

'Other sectors' include agriculture, commercial and public service, and residential.

porous rock 透水岩 non-porous rock 不透水岩 petrochemical industry 石油化學工業 petrochemical 石油化學產品
pesticide 殺蟲劑 pharmaceutical 藥物 cosmetic 化妝品 synthetic 合成的 polymer 聚合物

Fig. 25.6 shows products made from petroleum.



(a) CD cases



(b) Cleansers



(c) Food container

Fig. 25.6 Products made from petroleum

Economic importance of petroleum

According to Table 25.1, the majority of the world's petroleum supplies are found in politically sensitive areas of the world. The supply of petroleum and its price affect the whole global economy. Oil prices directly affect the prices of petrochemicals. Indirectly, oil prices affect costs such as transportation, manufacturing and heating. The increase in these costs can in turn affect the prices of a variety of goods and services, as producers may pass the production costs on to consumers.

Proved reserves are estimated to be worth-extracting with present technology and prices.

Table 25.1

Top ten oil reserve countries in the world

Rank	Country	Proved reserves (billion barrels)
1	Saudi Arabia	264.6
2	Venezula	209.4
3	Canada	173.6
4	Iran	151.2
5	Iraq	143.1
6	Kuwait	101.5
7	United Arab Emirates	97.8
8	Russia	60.0
9	Libya	48.1
10	Nigeria	38.5

(Source: US Central Intelligence Agency The World Factbook, 2012)

Fig. 25.7 shows the monthly average oil prices from 1990 to 2012. We can see that the price of oil quadrupled over the last decade.

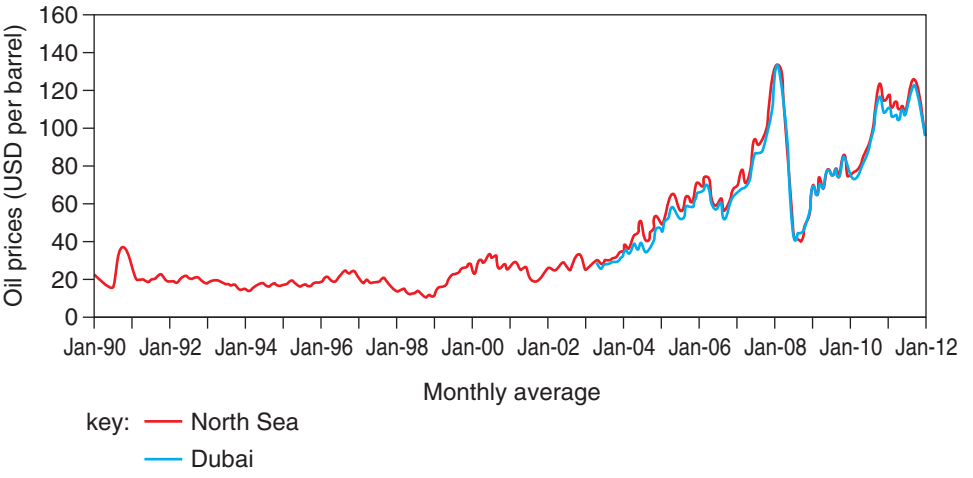


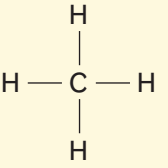
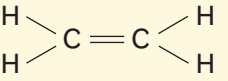
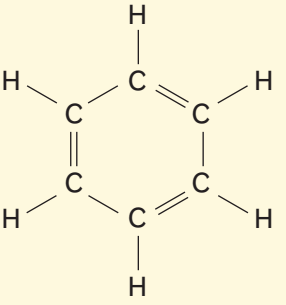
Fig. 25.7 Monthly average crude oil prices from 1990 to 2012 (Source: Dow Jones & Company)

25.3 Hydrocarbons

Hydrocarbons are compounds that contain only carbon and hydrogen atoms. There are different types of hydrocarbons. Table 25.2 shows three examples.

Table 25.2

Three examples of hydrocarbon

Name	Structure
Methane	
Ethene	
Benzene	

These diagrams show the structural formulae of the hydrocarbons. The structural formula of a compound shows how atoms are linked together by covalent bonds. We will discuss structural formulae in Unit 26.

hydrocarbon 烴（碳氫化合物）

The ring of six carbon atoms in **benzene** has special properties. Hydrocarbons that contain a benzene ring are called **aromatic hydrocarbons**. Hydrocarbons that do not contain a benzene ring are called **aliphatic hydrocarbons**.

25.4 What does petroleum contain?

Petroleum is a mixture of about 150 different hydrocarbons (Fig. 25.8). A small number of compounds containing nitrogen, oxygen and sulphur are also found in petroleum.

Percentage by mass of the most common elements in a representative sample of petroleum:

Carbon	83.0–87.0%
Hydrogen	10.0–14.0%
Nitrogen	0.1–2.0%
Oxygen	0.05–1.5%
Sulphur	0.05–6.0%



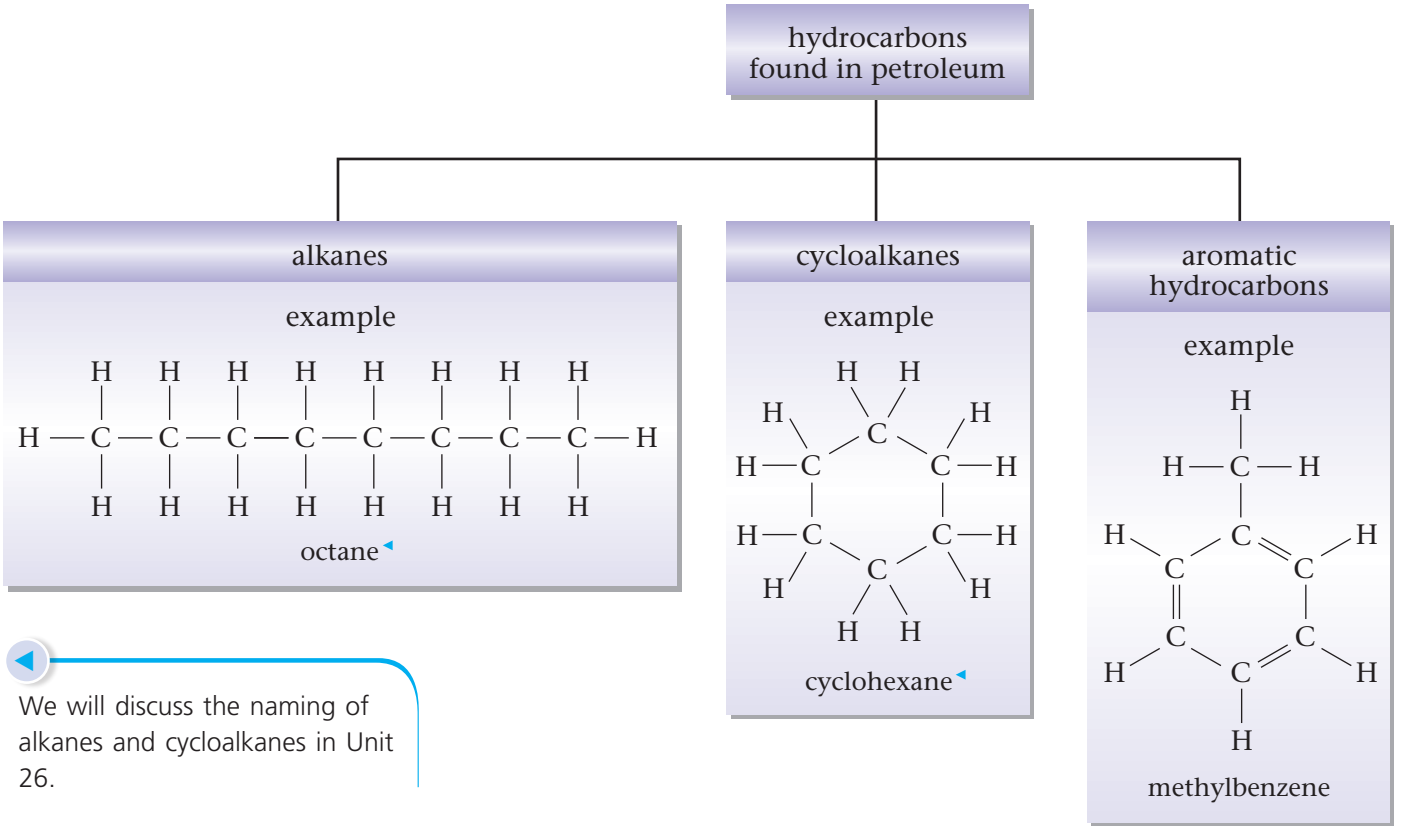
Fig. 25.8 Petroleum is a mixture of hydrocarbons

We can divide the hydrocarbons found in petroleum into three major classes of compounds:

- **alkanes**;
- **cycloalkanes**; and
- aromatic hydrocarbons.

benzene 苯 aromatic hydrocarbon 芳香族烴 aliphatic hydrocarbon 脂肪族烴 alkane 烷 cycloalkane 環烷

Alkanes are linear or branched aliphatic hydrocarbons that contain no double or triple bonds. Cycloalkanes are cyclic hydrocarbons that contain no double or triple bonds. Most cycloalkanes found in petroleum are based on five- or six-membered carbon rings (Fig. 25.9).



We will discuss the naming of alkanes and cycloalkanes in Unit 26.

Fig. 25.9 Classification of hydrocarbons found in petroleum

25.5 Fractional distillation of petroleum

To convert petroleum into useful products, we have to separate the mixture. We do this by heating it and collecting the **fractions** that boil over different temperature ranges. Each fraction is a mixture of hydrocarbons. The process is called **fractional distillation**.

In practice, fractional distillation of petroleum is carried out in an **oil refinery** (Fig. 25.10a). In an oil refinery, there are tall towers called **fractionating towers** (Fig. 25.10b).

DSE 2014 Paper 1B Q6(a)(i)

fraction 餾分　fractional distillation 分餾　oil refinery 煉油廠　fractionating tower 分餾塔



Fig. 25.10a An oil refinery



Fig. 25.10b Fractionating towers in an oil refinery

Animation
Fractional distillation of petroleum

Reference Website
Refer to an animation on the fractional distillation of petroleum in industry:
- http://ibchem.com/IB/ibnotes/full/ope_htm/oil_refining.htm (accessed August 2014)
- http://www.howstuffworks.com/environmental/energy/oil-refining4.htm (accessed August 2014)

Fractional distillation of petroleum involves the following processes:

- First heat the petroleum to about 400 °C in a furnace (Fig. 25.11). Compounds with higher boiling points remain as liquid while those with lower boiling points vaporize. Thus, petroleum is turned into a mixture of liquid and vapour.
- Pass the mixture of liquid and vapour into a fractionating tower that is cooler at the top than at the bottom.
- The liquid flows to the bottom of the tower and is called the residue. The residue contains hydrocarbons with over 25 carbon atoms per molecule.
- The vapour cools down as it rises through the tower.
- When a substance in the vapour arrives at a height where the temperature of the tower is lower than the boiling point of that substance, it will condense to form a liquid.
- The trays collect the various liquid fractions. The liquid fraction that condenses on each tray is piped off.
- Hydrocarbons with smaller molecules have lower boiling points. Hence they condense in the trays nearer to the top of the tower, where it is cooler.

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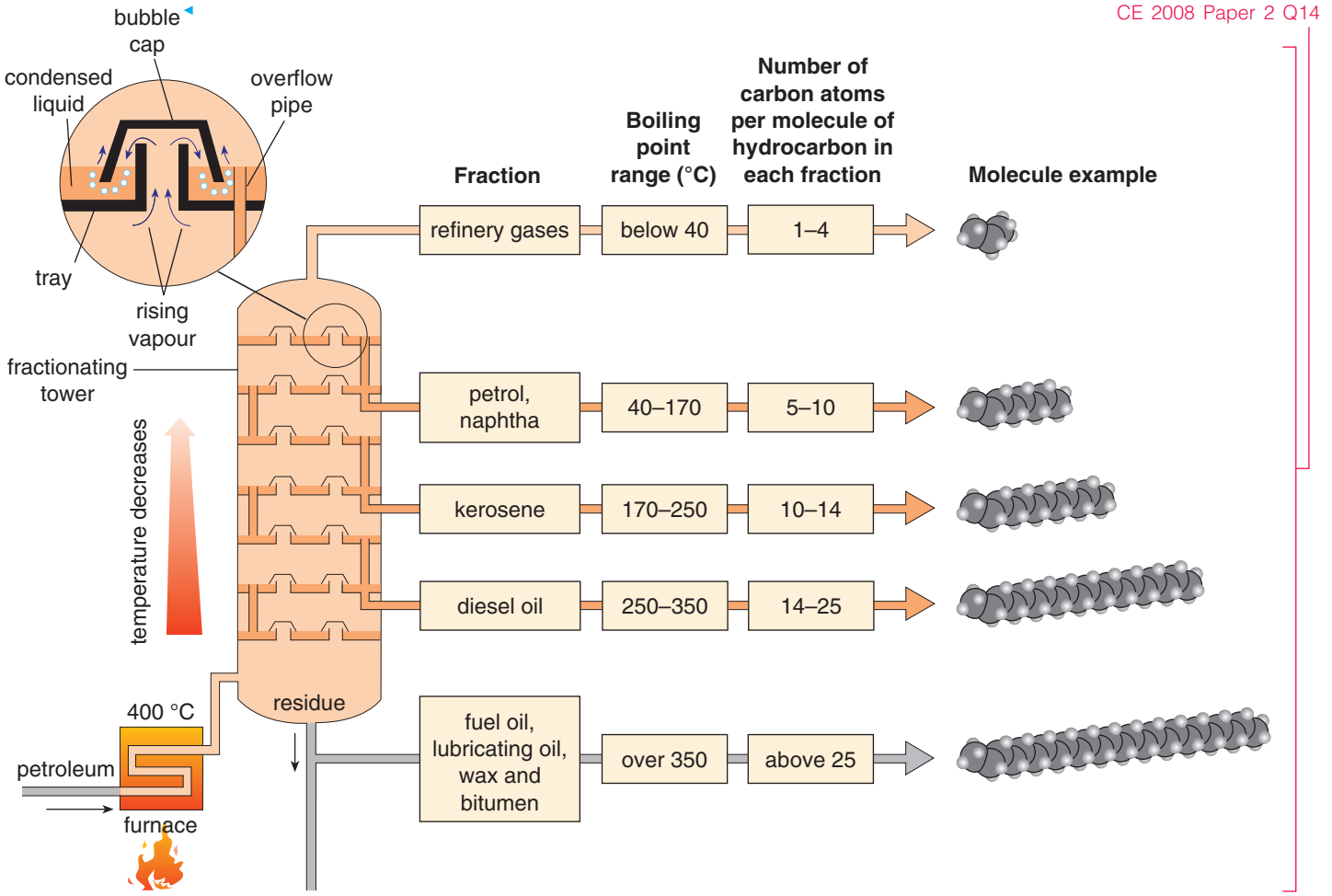


Fig. 25.11 Fractional distillation of petroleum in a fractionating tower

The vapour will bubble out and keep rising if its condensation point is lower than the temperature of the liquid in the tray.

Video
Fractional distillation of crude oil

25.1
Fractional distillation of crude oil and studying the properties of its fractions.

Fractional distillation of crude oil in the laboratory

We can carry out the fractional distillation of crude oil in the laboratory using the experimental set-up shown in Fig. 25.12.

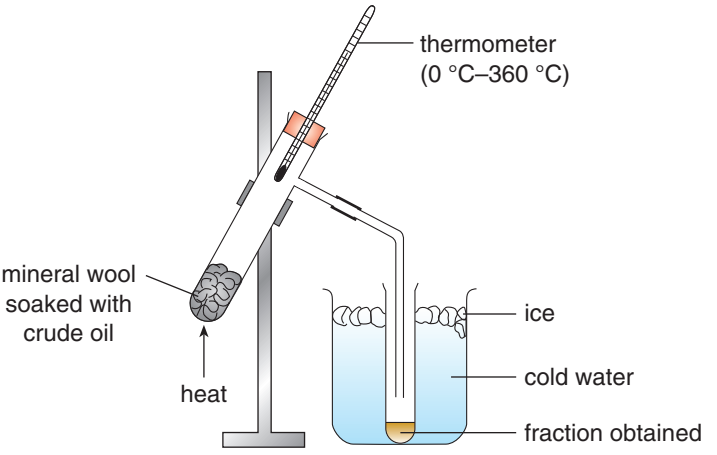


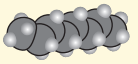
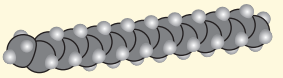
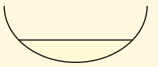
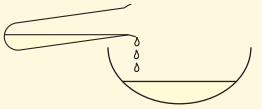
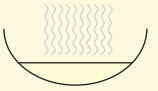
Fig. 25.12 Fractional distillation of crude oil in the laboratory

A fraction with a lower boiling point range contains hydrocarbons with a smaller number of carbon atoms in their molecules. A fraction with a higher boiling point range contains hydrocarbons with a greater number of carbon atoms in their molecules.

Table 25.3 compares the properties of a fraction with a lower boiling point range with those of a fraction with a higher boiling point range.

Table 25.3

Comparing the properties of a fraction with a lower boiling point range with those of a fraction with a higher boiling point range

	Fraction with a lower boiling point range	Fraction with a higher boiling point range
Number of carbon atoms per molecule	 smaller number of carbon atoms per molecule	 greater number of carbon atoms per molecule
Colour	 lighter	 darker
Viscosity	 less **viscous** (or more runny)	 more viscous (or thicker)
Ease of evaporation	 easier to evaporate (or more **volatile**[1])	 more difficult to evaporate (or less volatile)
Flammability	 more flammable; burns with a clearer flame	 less flammable; burns with a sootier flame

Reference Website

- Refer to an animation demonstrating the viscosity of different petroleum fractions: http://www.absorblearning.com/media/item.action?quick=11m (accessed August 2014)
- Refer to an animation illustrating the flammability of different petroleum fractions: http://www.absorblearning.com/media/item.action?quick=11n (accessed August 2014)

Do you know

Getting the volatility right

A car is moved by burning a mixture of air and petrol vapour in an engine. High volatility is a crucial property for petrol to burn in an engine. On extremely cold winter mornings, the petrol must be volatile enough to vaporize and mix with air. Otherwise, the engine would not start. However, on hot summer days, if the petrol is too volatile, it may vaporize in the fuel feed-pipe and so cause a vapour lock that prevents the petrol from mixing with air and reaching the engine. Again, the engine would not start.

So, in the winter petrol is made to contain more of the alkanes with low boiling points, while in the summer more compounds of low volatility are incorporated.

[1] Volatility is a measure of how easily molecules escape from a liquid. A volatile liquid loses molecules from the surface very easily.

viscous 黏滯　　volatile 易揮發的

25.6 Major uses of fractions of petroleum

The major uses of a petroleum fraction are closely related to its properties. For example, refinery gases contain hydrocarbons with a small number of carbon atoms in their molecules. They ignite easily and are good fuels. They can be piped easily to where they are needed. However, refinery gases are bulky to store, so they are often stored under pressure as liquids.

LPG is liquefied petroleum gas, a common domestic fuel in Hong Kong.

Fig. 25.13 shows the names, properties and major uses of the main fractions of petroleum.

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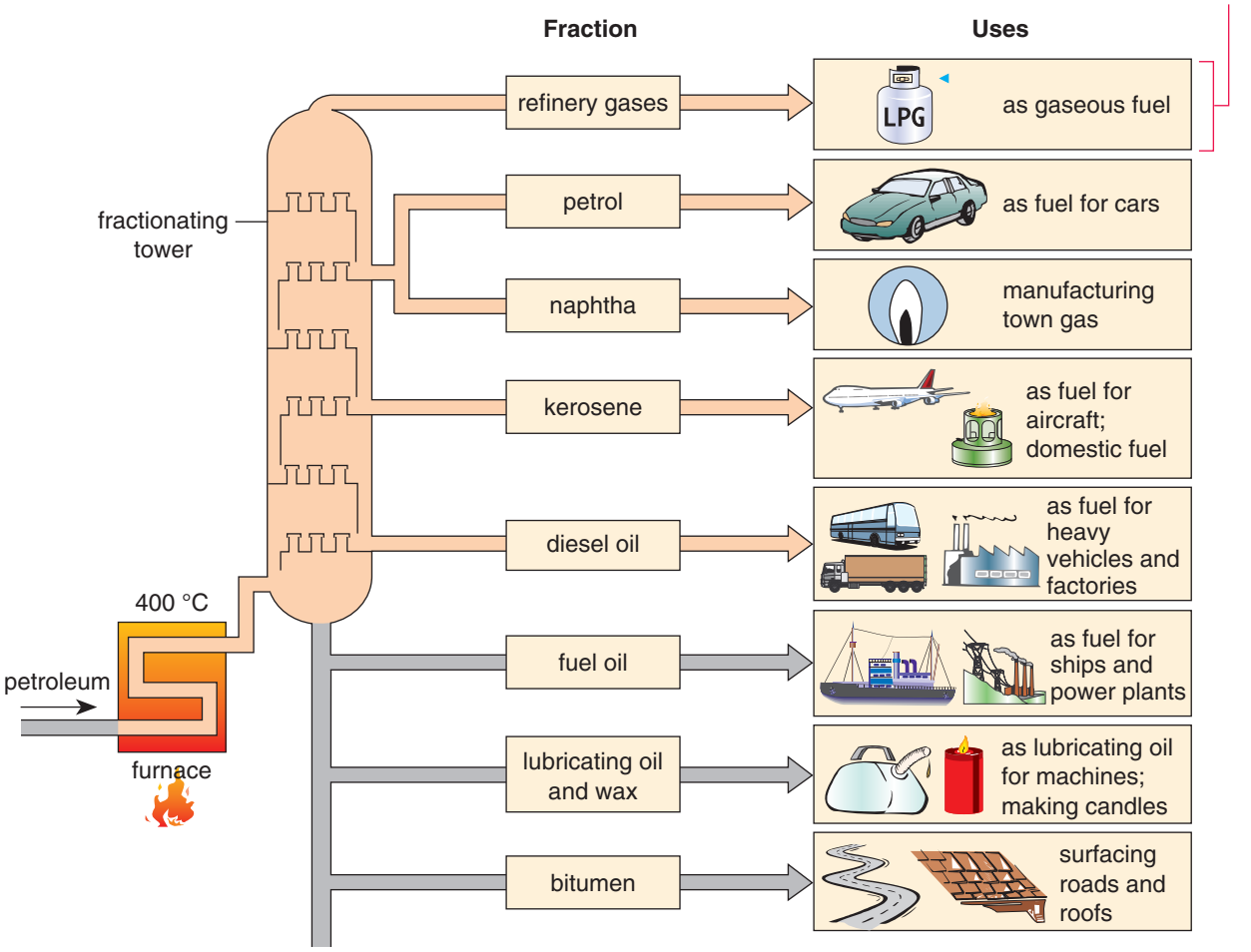


Fig. 25.13 Names and major uses of the main fractions of petroleum

refinery gas 煉油氣　petrol 汽油　naphtha 石腦油　kerosene 煤油　diesel oil 柴油　fuel oil 燃料油
lubricating oil 潤滑油　wax 蠟　bitumen 瀝青



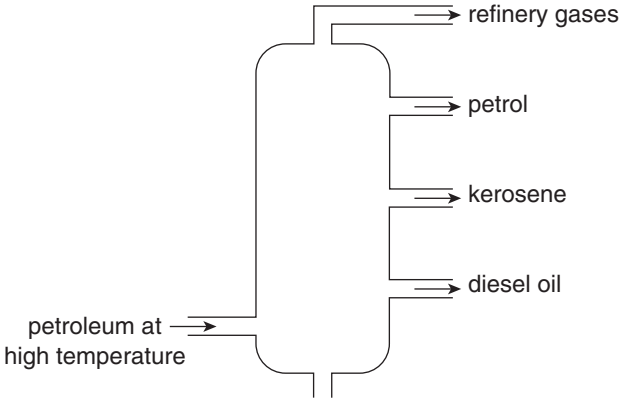
Fig. 25.14 Kerosene is used as a fuel for aircraft



Fig. 25.15 Bitumen is used for surfacing roads

Practice 25.1

The diagram below shows a tower used to separate petroleum into fractions.



a) Name the process that is used to separate the petroleum into fractions.
Fractional distillation

b) Which of the four fractions has the LOWEST boiling point range?
Refinery gas

c) Which of the four fractions is the most viscous?
Diesel oil

d) Which of the four fractions is used as a fuel for aircraft?
Kerosene

e) Give ONE reason to account for the differences in properties of fractions at the different outlets.
Hydrocarbons in different fractions contain different number of carbon atoms.

25.7 Combustion of hydrocarbons

During combustion, a fuel reacts with oxygen. The oxygen usually comes from the air. Petroleum is composed mainly of hydrocarbons. When a hydrocarbon burns completely, it is oxidized to form carbon dioxide and water. Take the combustion of octane, a hydrocarbon, as an example:

$$\underset{\text{octane}}{2C_8H_{18}(g)} + 25O_2(g) \longrightarrow 16CO_2(g) + 18H_2O(l)$$

Combustion is a gas-phase reaction. Thus, liquid and solid alkanes must first be vaporized.

Exothermic and endothermic reactions

We will discuss exothermic and endothermic reactions in Topic 9 Chemical Reactions and Energy.

Combustion of hydrocarbons releases a great deal of heat. Reactions in which heat is released are called **exothermic reactions** (Fig. 25.16). We will notice a temperature rise in such reactions. Some reactions take in heat. They are **endothermic reactions** (Fig. 25.17). For example, heat is needed for the decomposition of calcium carbonate.

$$CaCO_3(s) \xrightarrow{\text{heat}} CaO(s) + CO_2(g)$$

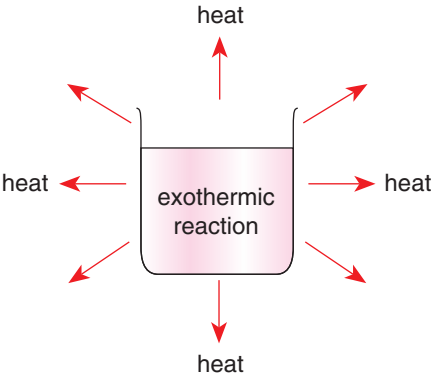


Fig. 25.16 Heat is released during an exothermic reaction

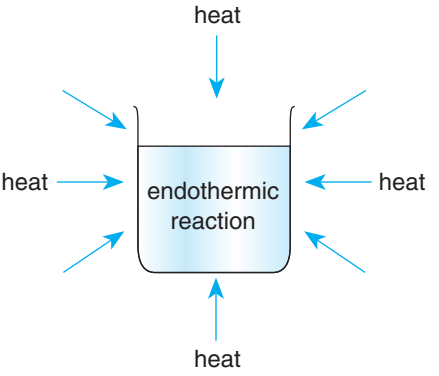


Fig. 25.17 Heat is taken in during an endothermic reaction

✓ An exothermic reaction is a reaction in which heat is released. An endothermic reaction is a reaction in which heat is taken in.

exothermic reaction 放熱反應　　endothermic reaction 吸熱反應

Video
Greenhouse effect

25.8 Carbon dioxide and the greenhouse effect

Burning of fossil fuels produces carbon dioxide and water vapour. Both of these are referred to as **greenhouse gases**, but what are greenhouse gases and how did they get their name?

Fig. 25.18 shows a greenhouse. The glass of the greenhouse allows sunlight to pass through, but prevents most of the **infrared radiation** from escaping. The heat trapped by the glass walls warms the air inside.

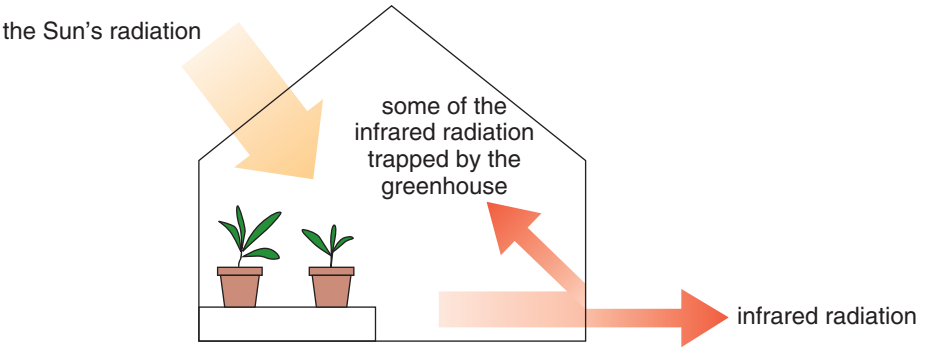


Fig. 25.18 In a greenhouse, the heat trapped by the glass walls warms the air inside

The Earth's atmosphere reflects about 30% of the Sun's radiation back into space. The remaining 70% passes through the atmosphere to the surface of the Earth. The Earth absorbs this radiation and also radiates its own infrared radiation.

Greenhouse gases in the atmosphere behave like glass of a greenhouse. They trap some of the infrared radiation which is radiated from the Earth, thus keeping the atmosphere warm for life to sustain on Earth. This heat-trapping feature of the atmosphere is known as the **greenhouse effect** (Fig. 25.19). N5

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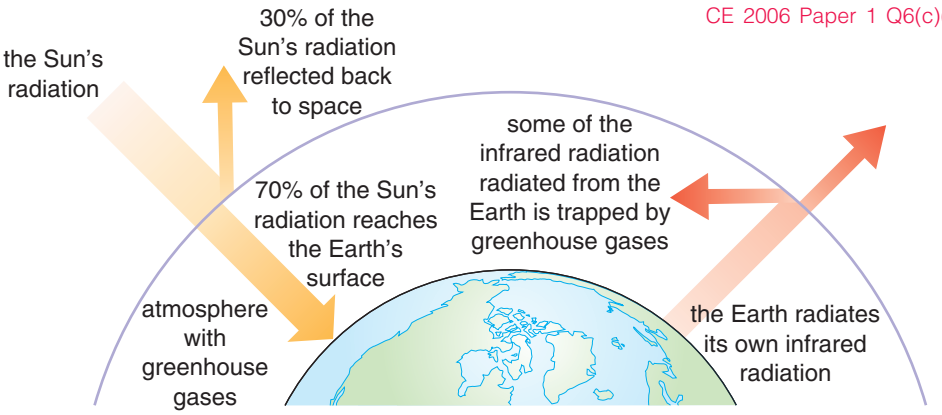


Fig. 25.19 Greenhouse gases trap some of the infrared radiation, keeping the atmosphere warm

N5 Sources of greenhouse gases

The following table shows the sources of greenhouse gases.

Greenhouse gas	Source(s)
Carbon dioxide	burning fossil fuels and forests
Methane	swamps, paddy fields, animal dung, rubbish tips
Chlorofluorocarbons (CFCs)	refrigerators, aerosol sprays
Oxides of nitrogen	burning fossil fuels and forests
Ozone	ozone layer

Some greenhouse gases have a greater greenhouse effect than others. The greenhouse factor can be used to compare the greenhouse effect of the different gases. The measure is relative to the same amount of carbon dioxide that is assigned the value of 1.

Gas	Percentage in troposphere	Greenhouse factor
CO_2	0.035	1
N_2O	0.00003	160
CH_4	0.00017	30
CCl_3F	2.8×10^{-8}	21 000
CCl_2F_2	4.8×10^{-6}	25 000
$H_2O(g)$	1	0.1
O_3	4×10^{-6}	2 000
O_2	20	negligible
N_2	78	negligible
Ar	1	negligible

greenhouse gas 温室氣體　　infrared radiation 紅外光　　greenhouse effect 温室效應

CE 2006 Paper 1 Q6(a)&(c)(ii)

Carbon dioxide is the most important greenhouse gas. Over the past hundred years or so, we have burnt large amounts of fossil fuels due to various human activities, such as transportation, industrial production, electricity generation, etc. This has put much extra carbon dioxide into the atmosphere. N6

Fig. 25.20a shows the approximate carbon dioxide level in the atmosphere from 1900 to 2012. The level of carbon dioxide has been rising. Gradually, the Earth's temperature has been rising too (Fig. 25.20b). A majority of scientists believes that the increasing level of carbon dioxide is the cause of **global warming**. This global warming could lead to climate change and rising sea levels, which could cause flooding (Fig. 25.21).

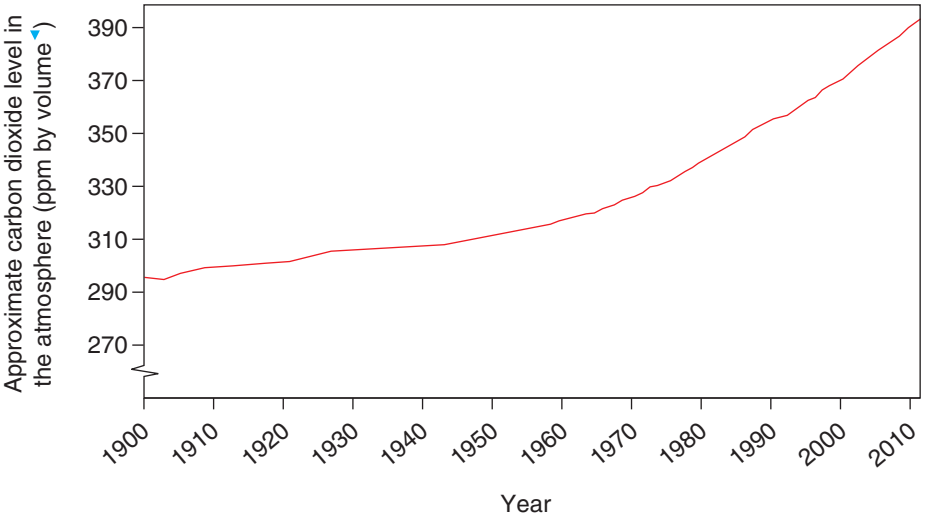


Fig. 25.20a The approximate carbon dioxide level in the atmosphere from 1900 to 2012 (Source: US National Oceanic and Atmospheric Administration Earth System Research Laboratory)

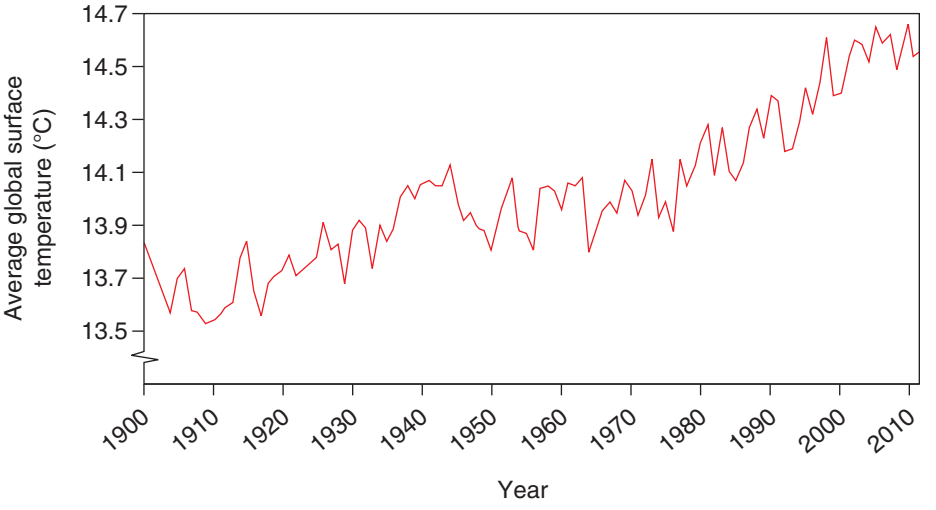


Fig. 25.20b Average global surface temperature from 1900 to 2012 (Source: US National Aeronautics and Space Administration Goddard Institute for Space Studies)

Do you know

Methane as a greenhouse gas

Carbon dioxide and water vapour are not the only greenhouse gases. Methane is one too. It is released as a result of extracting crude oil, natural gas and coal. Large amounts of methane are also produced by cattle and rice paddy fields. Releasing methane to the atmosphere is about 20 times more effective in increasing global temperature than releasing equal amounts of carbon dioxide.

ppm by volume = volume (in cm^3) of carbon dioxide in 10^6 cm^3 of air

N6 Although carbon dioxide is a greenhouse gas, it is NOT considered as an air pollutant.
Carbon dioxide does NOT lead to the formation of acid rain.

global warming 全球增温

Practice 25.2

a) The Earth absorbs the Sun's radiation and also radiates its own infrared radiation.
Greenhouse gases trap some of the infrared radiation which is radiated from the Earth, thus keeping the atmosphere warm for life to sustain on Earth.

b) Any one of the following:
- Burning fossil fuels
- Production of cement
- Making iron / steel
- Deforestation
- Fermentation
- Oil refining
- Respiration

c) Any one of the following:
- The remains / manure of cattles decay to give methane.
- Cattles discharge methane in belches from partly digested food in their guts.
- Action of anaerobic bacteria in environments lacking of oxygen such as wetlands and paddy rice fields.

d) Any two of the following:
- Burn a fuel from a plant source (e.g. bioethanol). Burning the fuel returns carbon dioxide which has been removed from the atmosphere during the photosynthesis.
- Use alternative energy source (e.g. solar energy, wind energy, nuclear, hydroelectric power, geothermal energy)
- Improve the efficiency of the process. / Use a catalyst so that the process requires less activation energy.



Fig. 25.21 Climate change and rising sea levels will cause flooding

Do you know ?

International cooperation to fight against global warming

Having recognized that global climate change could result from increasing concentrations of greenhouse gases in the atmosphere, the *World Meteorological Organization (WMO)* and the *United Nations Environment Programme (UNEP)* established the *Intergovernmental Panel on Climate Change (IPCC)* in 1988. This organization assesses all the evidence from thousands of scientists doing research about climate change. Its report says that it is 90% certain that global warming and climate change are caused by human activities.

The *Kyoto Protocol* was the first international attempt at curbing greenhouse gas emissions. It was adopted in 1997.

Practice 25.2

Carbon dioxide and methane are major greenhouse gases in the atmosphere. There was a large increase in the concentration of both gases in the atmosphere in the past 100 years.

a) State the importance of the greenhouse gases in the atmosphere to living things on Earth. — CE 2006 Paper 1 Q6(c)(i)

b) Suggest ONE process by which carbon dioxide is added to the atmosphere. — CE 2006 Paper 1 Q6(a)&(b)

c) Suggest ONE process by which methane is added to the atmosphere.

d) Most carbon dioxide from industrial processes is allowed to escape into the atmosphere. CE 2006 Paper 1 Q6(c)(iii)(I)

Suggest TWO changes that a chemical company could make to its processes to achieve a reduction in the rate of increase of carbon dioxide levels.

World Meteorological Organization (WMO) 世界氣象組織 United Nations Environment Programme (UNEP) 聯合國環境規劃署
Intergovernmental Panel on Climate Change (IPCC) 政府間氣候變化專門委員會 Kyoto Protocol 京都議定書



Fig. 25.22 Poor air quality affects our everyday lives

25.9 Major air pollutants from cars, factories, incinerators and power plants N7

Most air pollution comes from the burning of fossil fuels in car engines, factories, *incinerators* and power plants. Besides carbon dioxide, harmful substances such as carbon monoxide, unburnt hydrocarbons, **suspended particulates**, **oxides of nitrogen** and sulphur dioxide are also emitted into the atmosphere (Fig. 25.22). N8

Carbon monoxide

Sometimes a hydrocarbon fuel◂ cannot get enough oxygen to burn completely. As a result, incomplete combustion of the fuel occurs. The carbon in the fuel forms carbon monoxide instead of carbon dioxide.

Most of the carbon monoxide in the air comes from the exhaust gas of cars, trucks and buses.

ASL 2005 Q11(a)(i)(1)

◂ For example, petrol in a car engine may not get enough oxygen to burn completely.

N7 Examination questions often ask about the current air pollution problems in Hong Kong and possible measures for reducing the emission of air pollutants.

N8 Refer to page T16.

Video
Major air pollutants and their effects on our health

Unburnt hydrocarbons

Not all petrol in car engines gets burnt. Some unburnt hydrocarbons are released into the surrounding environment. Also, volatile hydrocarbons, a kind of **volatile organic compounds** (VOCs), may evaporate from the petrol tank into the air. These unburnt hydrocarbons are the agents for various urban pollution problems and also one of the potential causes of cancer.



Fig. 25.23 Dark smoke from a motor vehicle

Suspended particulates

Incomplete combustion of hydrocarbons produces dark smoke which contains mainly carbon particles (Fig. 25.23). These small particles remain suspended in the air for a long time and are called suspended particulates. Some carbon particles are too small to be filtered by hairs in the nose◂. They will irritate our respiratory system.

Reference Website
Refer to the video clip 'Environment and sustainable development'（環境與可持續發展）:
http://resources.hkedcity.net/resource_detail.php?rid=345340988
(accessed August 2014)

◂ Such carbon particles are called **respirable suspended particulates** (RSP).

incinerator 焚化爐　suspended particulate 懸浮粒子　oxides of nitrogen 氮氧化物
volatile organic compound 揮發性有機化合物　respirable suspended particulate 可吸入的懸浮粒子

Oxides of nitrogen

Nitrogen and oxygen in the air do not normally react. Inside car engines and power plant furnaces, the air gets so hot and it causes nitrogen and oxygen to react, forming oxides of nitrogen (NO_x) — mainly nitrogen monoxide.

Nitrogen monoxide can be further oxidized in the air to nitrogen dioxide.

$$N_2(g) + O_2(g) \longrightarrow 2NO(g)$$
nitrogen monoxide

$$2NO(g) + O_2(g) \longrightarrow 2NO_2(g)$$
nitrogen dioxide

ASL 2005 Q11(a)(i)(2)

Sulphur dioxide

Factories and power plants burn either coal or low-grade petroleum, both containing sulphur. When these fuels are burnt, they emit sulphur dioxide into the air.

Incinerators burn a lot of rubbish every day (Fig. 25.24), much of it being paper containing sulphur compounds. Hence the smoke from incinerators contains large amounts of sulphur dioxide.

There is no large-scale rubbish incinerator in Hong Kong. Most solid waste is buried by landfilling.

Air pollutants such as nitrogen dioxide and sulphur dioxide react with rainwater to form acids that lower the pH of rainwater. This gives rise to acid rain. N9

N9 Examination questions often ask about acid rain:
- air pollutants that lead to acid rain (NO_x and SO_2);
- environmental problems caused by acid rain.

$$2NO_2(g) + H_2O(l) \longrightarrow HNO_3(aq) + HNO_2(aq)$$
nitric acid nitrous acid

$$SO_2(g) + H_2O(l) \rightleftharpoons H_2SO_3(aq)$$
sulphurous acid

CE 2008 Paper 2 Q1



Fig. 25.24 A waste incinerator

Fig. 25.25 summarizes the effects of the air pollutants mentioned above.

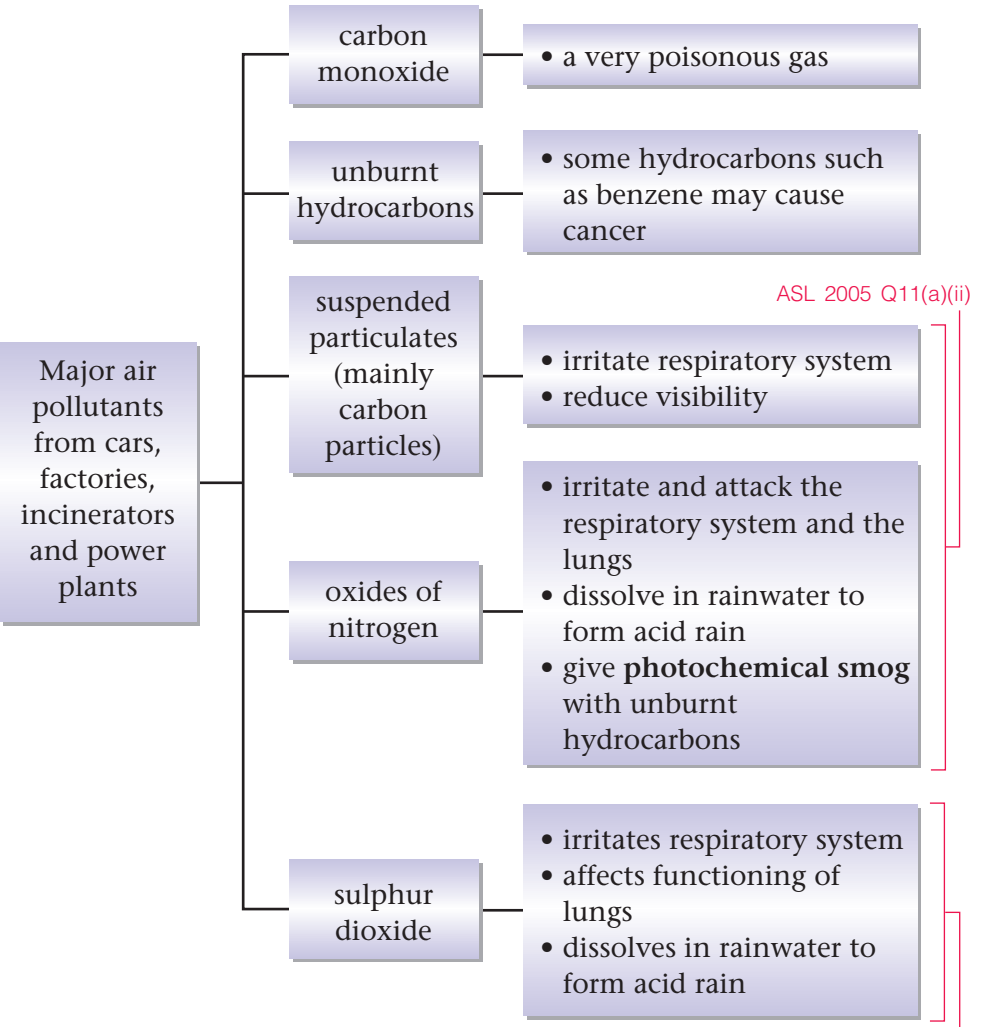


Fig. 25.25 The effects of major air pollutants



Fig. 25.26 This forest was badly damaged by acid rain

Do you know

Effects of acid rain

Many buildings, particularly those made of limestone and marble, are damaged by acid rain.

Acid rain harms plants that take in the acidic water and the animals that live in the affected rivers and lakes. Calcium and magnesium ions are washed out of the soil by acid rain, depleting the minerals available to plants. Acid rain also washes aluminium ions out of the soil and releases them into rivers and lakes. These ions are toxic to fish.

photochemical smog 光化煙霧



Fig. 25.27 Photochemical smog over Los Angeles, US

Do you know

How do oxides of nitrogen and unburnt hydrocarbons cause photochemical smog?

Photochemical smog is a mixture of pollutants formed by the reaction of oxides of nitrogen and unburnt hydrocarbons under the influence of sunlight. Pollutants in photochemical smog include particulates, nitrogen oxides, ozone and hydrocarbons, etc.

The chemistry of the formation of photochemical smog is quite complex. Firstly, nitrogen dioxide molecules absorb sunlight and break apart to form nitrogen monoxide molecules and free oxygen atoms which are very reactive.

$$NO_2(g) \xrightarrow{\text{sunlight}} NO(g) + O(g)$$

The oxygen atoms then react with unburnt hydrocarbons and other components of the atmosphere (e.g. oxygen) to produce a variety of irritating and toxic chemicals.

25.10 Measures for reducing emission of air pollutants

Table 25.4 summarizes possible measures for reducing the emission of air pollutants from the combustion of fossil fuels. N10

Table 25.4

Possible measures for reducing the emission of air pollutants from the combustion of fossil fuels

Pollutant	Measures for reducing the emission of the pollutant
Oxides of nitrogen	• installing **catalytic converters** in exhaust systems of motor vehicles • installing low nitrogen oxide burners in power plants
Sulphur dioxide	• using low-sulphur coal instead of high-sulphur coal N11 • installing flue gas **desulphurization systems** or **scrubbers** in power plants N12
Particulates	• using **electrostatic precipitators** in power plants N13 • installing **particulate removal devices** in motor vehicles

DSE 2014 Paper 1B Q6(b)(ii)
CE 2006 Paper 2 Q22
CE 2010 Paper 2 Q25
CE 2011 Paper 1 Q6(c)
CE 2007 Paper 2 Q26
DSE 2012 Paper 1B Q10

N10 The following measures can help reduce the level of carbon monoxide at the road side:
- using liquefied petroleum gas as fuel for motor vehicles;
- installing catalytic converters for motor vehicles.

N11 Advantages of using natural gas over using coal as a fuel in power stations:
- natural gas burns more completely;
- natural gas has less sulphur-containing substances.

N12 Refer to page T16.

N13 Refer to page T17.

catalytic converter 催化轉化器 desulphurization system 脱硫系統 scrubber 滌氣器 electrostatic precipitator 靜電沉積器 particulate removal device 微粒消減裝置



Fig. 25.28 A LPG public light bus

Air pollution in Hong Kong

Hong Kong's air pollution is mainly contributed by motor vehicles and power plants. Diesel vehicles, particularly trucks, buses and light buses, are the main source of street-level air pollution.

To tackle this problem, the Government has implemented various measures, such as the incentive programme to replace diesel taxis / light buses with liquefied petroleum gas vehicles (Fig. 25.28), the adoption of tighter vehicle emission standards, and the reduction of first registration tax for environment-friendly vehicles.

The power plants in Hong Kong use mainly coal and natural gas as fuels for electricity generation. Compared with coal, natural gas is a cleaner fuel. In recent years, the Government has urged the power companies to increase the use of natural gas for electricity generation.

Figs. 25.29 and 25.30 show the emissions of oxides of nitrogen and respirable suspended particulates in Hong Kong (1997–2011). The emissions have declined over the years.

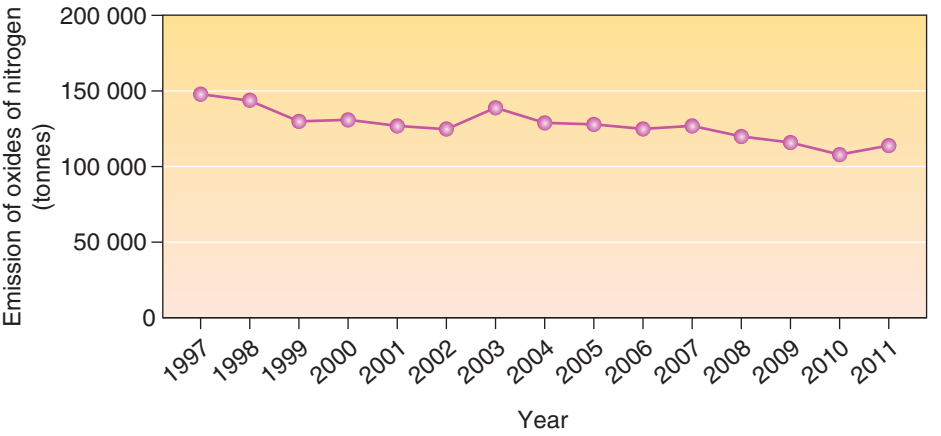


Fig. 25.29 Emission of oxides of nitrogen in Hong Kong, 1997–2011
(Source: Environmental Protection Department)

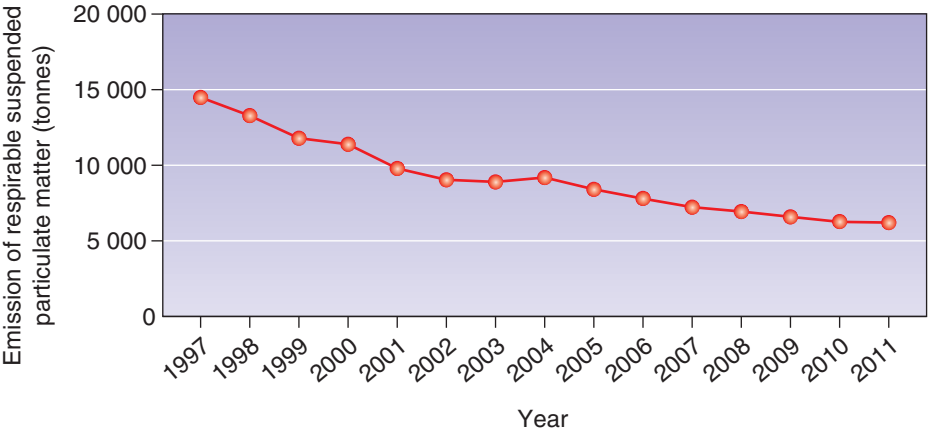


Fig. 25.30 Emission of respirable suspended particulates in Hong Kong, 1997–2011
(Source: Environmental Protection Department)

Using catalytic converters in exhaust systems of motor vehicles

Catalytic converters are fitted to the exhaust systems of motor vehicles to reduce the emission of carbon monoxide, oxides of nitrogen and unburnt hydrocarbons.

A catalytic converter (Fig. 25.31) is composed of a metal casing that has a ceramic honeycomb-shaped interior coated with platinum and rhodium metals. These metals act as **catalysts**.

A catalyst is a substance which alters the rate of a reaction without itself undergoing any permanent chemical changes. We will discuss catalysts in Topic 10 Rate of Reaction.

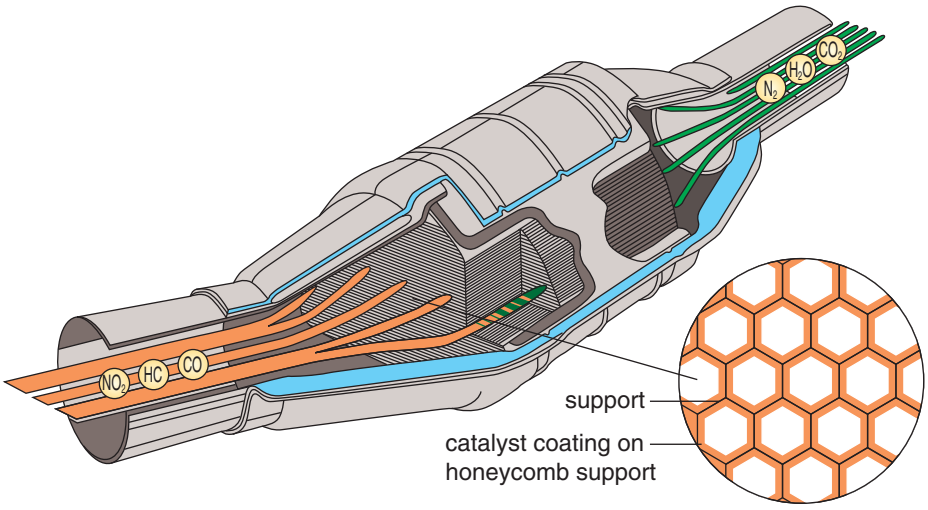


Fig. 25.31 Air pollutants in exhaust gas are turned into relatively harmless substances after passing through a catalytic converter

The honeycomb-shaped interior of the converter provides the maximum surface area on which reactions can take place, while using the least amount of catalysts.

As the exhaust gas passes over the catalysts, an oxidation-reduction reaction occurs. Carbon monoxide is oxidized to carbon dioxide. Nitrogen oxides are reduced to nitrogen. N14

$$2CO(g) + 2NO(g) \xrightarrow{\text{catalyst}} 2CO_2(g) + N_2(g)$$

The excess carbon monoxide and unburnt hydrocarbons are oxidized by air to carbon dioxide and water. N15

$$2CO(g) + O_2(g) \xrightarrow{\text{catalyst}} 2CO_2(g)$$

$$2C_8H_{18}(l) + 25O_2(g) \xrightarrow{\text{catalyst}} 16CO_2(g) + 18H_2O(l)$$

octane

N14 A catalytic converter is NOT used to remove sulphur dioxide. Be careful with the spelling of the term 'catalytic converter'.

N15 The following environmental problems may be reduced by installing catalytic converters in motor vehicles:
- acid rain;
- photochemical smog.

Greenhouse effect CANNOT be reduced as the emission of carbon dioxide gas is increased by installing catalytic converters.

Octane is used to represent a typical hydrocarbon in petrol.

CE 2009 Paper 2 Q26

ASL 2005 Q11(a)(iii)
CE 2011 Paper 1 Q6(b)

catalyst 催化劑

Practice 25.3

1 Catalytic converters have been installed in most petrol-driven vehicles.

a) State the functions of a catalytic converter.

b) State ONE harmless product emitted from a catalytic converter.

c) Identify a metal used as a catalyst in a catalytic converter.

Suggest ONE reason, other than cost, why the catalyst is coated on a ceramic honeycomb.

2 a) Explain why power plants burning fossil fuels produce a considerable amount of sulphur dioxide.

b) Suggest TWO measures to reduce the emission of sulphur dioxide from such power plants.

Practice 25.3

1 a) Reduce the amount of
- carbon monoxide;
- unburnt hydrocarbons;
- oxides of nitrogen

in the exhaust gas.

b) Any one of the following:
- Nitrogen
- Water
- Carbon dioxide

c) Platinum / rhodium
Larger surface area available for reaction.

2 a) The fossil fuels used contain sulphur.
When the fuel burns, the sulphur is oxidized to sulphur dioxide.

b) Use low-sulphur coal.
Install flue gas desulphurization systems / scrubbers.

25.11 Impact of using fossil fuels on our quality of life and the environment

In our everyday life, energy is required to power vehicles and industries, to heat and cool our homes and workplaces, and to run appliances. Energy is fundamental to the quality of our lives. In fact, most modern human activities depend on the regular and uninterrupted supply of energy.

Most of the energy we use comes from the burning of fossil fuels. Hence these fuels have important economic impacts on our society. However, burning fossil fuels causes serious air pollution problems. It is crucial that the hazard posed by fossil fuels to the environment be tactfully minimized while taking into account its positive contribution to civilization.

Find & Share

Worksheet Assessment Form

A worksheet and an assessment form on this activity are available for download from *Jing Kung Chemistry Website*.

Impacts of using fossil fuels on our quality of life and the environment

Task

1 Conduct a library search on the topic.

2 Write a short essay of not more than 1 000 words on the topic.

Hints for the search

1 In what ways do you use energy from fossil fuels in a typical day?

2 What useful chemicals and products are produced from petroleum?

3 What are the environmental impacts of using fossil fuels? How does this affect you?

Refer to page T17.

Have you mastered?

Key terms

fossil fuel 化石燃料 2 coal 煤 2 petroleum 石油 2 natural gas 天然氣 2

petrochemical industry 石油化學工業 4 petrochemical 石油化學產品 4 polymer 聚合物 4

hydrocarbon 烴（碳氫化合物）6 benzene 苯 7 aromatic hydrocarbon 芳香族烴 7

aliphatic hydrocarbon 脂肪族烴 7 alkane 烷 7 cycloalkane 環烷 7 fraction 餾分 8

fractional distillation 分餾 8 oil refinery 煉油廠 8 fractionating tower 分餾塔 8

viscous 黏滯 11 volatile 易揮發的 11 refinery gas 煉油氣 12 petrol 汽油 12

naphtha 石腦油 12 kerosene 煤油 12 diesel oil 柴油 12 fuel oil 燃料油 12

lubricating oil 潤滑油 12 wax 蠟 12 bitumen 瀝青 12

exothermic reaction 放熱反應 14 endothermic reaction 吸熱反應 14 greenhouse gas 温室氣體 15

infrared radiation 紅外光 15 greenhouse effect 温室效應 15 global warming 全球增温 16

air pollutant 空氣污染物 18 incinerator 焚化爐 18 suspended particulate 懸浮粒子 18

oxides of nitrogen 氮氧化物 18 volatile organic compound 揮發性有機化合物 18

respirable suspended particulate 可吸入的懸浮粒子 18 acid rain 酸雨 20

photochemical smog 光化煙霧 21 catalytic converter 催化轉化器 21

desulphurization system 脱硫系統 21 scrubber 滌氣器 21

electrostatic precipitator 靜電沉積器 21 particulate removal device 微粒消減裝置 21

catalyst 催化劑 23

Checklist

After studying this unit, you should be able to

- ☐ describe the origin of fossil fuels;
- ☐ describe the industrial separation of petroleum into useful fractions by fractional distillation;
- ☐ relate the gradation in properties of the various fractions to the number of carbon atoms in the hydrocarbon molecules;
- ☐ describe the major uses of the various petroleum fractions;
- ☐ discuss the relationship between global warming and the use of fossil fuels;
- ☐ recognize the pollution from the combustion of fossil fuels;
- ☐ evaluate the impact of using fossil fuels on our quality of life and the environment;
- ☐ suggest measures for reducing the emission of air pollutants from the combustion of fossil fuels.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Have you mastered?

Summary

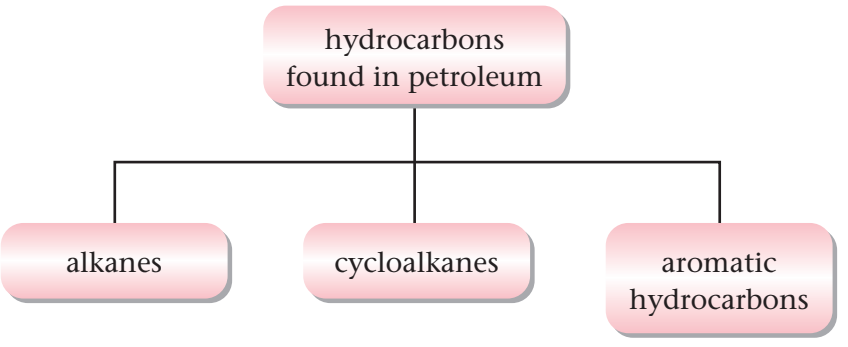
1 The major fossil fuels are coal, petroleum and natural gas.

2 Fossil fuels were formed from the remains of plants and animals that lived millions of years ago.

3 Hydrocarbons that contain a benzene ring are called aromatic hydrocarbons.

Hydrocarbons that do not contain a benzene ring are called aliphatic hydrocarbons.

4 Petroleum is a mixture mainly composed of hydrocarbons. Hydrocarbons are compounds that contain only carbon and hydrogen atoms.



5 Fractional distillation separates petroleum into fractions with different boiling point ranges.

Fraction	Boiling point range (°C)	Number of carbon atoms per molecule	General Properties
Refinery gases	below 40	1–4	fractions containing hydrocarbons with a smaller number of carbon atoms in their molecules: • lighter in colour • less viscous • easier to evaporate • more flammable • burn with a clearer flame
Petrol, naphtha	40–170	5–10	
Kerosene	170–250	10–14	
Diesel oil	250–350	14–25	fractions containing hydrocarbons with a greater number of carbon atoms in their molecules: • darker in colour • more viscous • more difficult to evaporate • less flammable • burn with a sootier flame
Fuel oil, lubricating oil and wax	over 350	above 25	

6 The main uses of refined petroleum are:

a) as fuels for transportation;

b) as fuels for generating electricity;

c) as fuels for heating;

d) as raw materials in the petrochemical industry.

7 Combustion of hydrocarbons is an exothermic reaction.

8 Burning of fossil fuels produces carbon dioxide and water vapour, which contribute to the greenhouse effect.

9 Major air pollutants from using fossil fuels include:

a) carbon monoxide;

b) unburnt hydrocarbons / volatile organic compounds;

c) suspended particulates;

d) oxides of nitrogen;

e) sulphur dioxide.

10 The table below summarizes possible measures for reducing the emission of air pollutants from the combustion of fossil fuels.

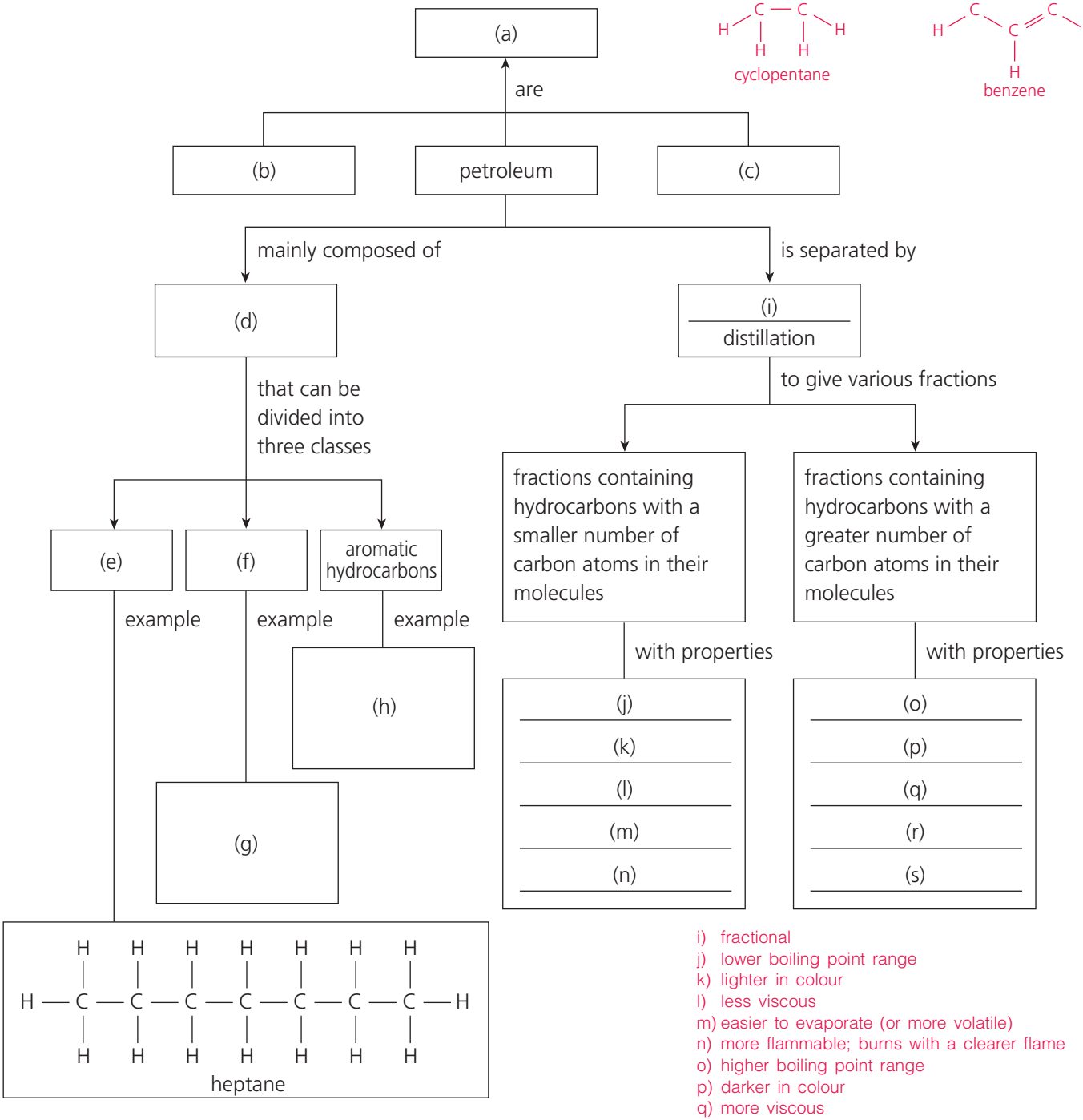
Pollutant	Measures for reducing the emission of the pollutant
Oxides of nitrogen	• installing catalytic converters in exhaust systems of motor vehicles • installing low nitrogen oxide burners in power plants
Sulphur dioxide	• using low-sulphur coal instead of high-sulphur coal • installing flue gas desulphurization systems or scrubbers in power plants
Particulates	• using electrostatic precipitators in power plants • installing particulate removal devices in motor vehicles

Unit Exercise

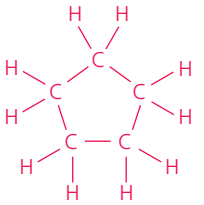
Note: The symbol ! indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Complete the following concept map.

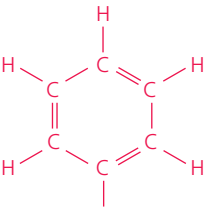


a) fossil fuels
b) coal
c) natural gas
d) hydrocarbons
e) alkanes
f) cycloalkanes
g)



cyclopentane

h)

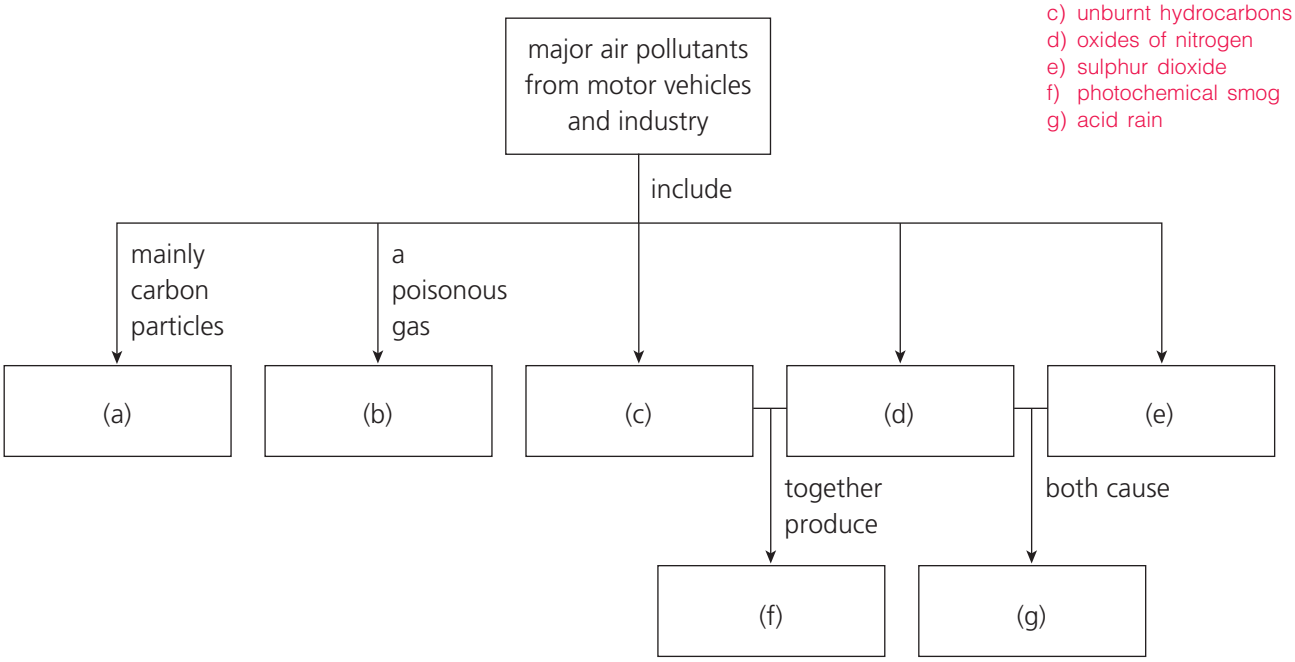


benzene

i) fractional
j) lower boiling point range
k) lighter in colour
l) less viscous
m) easier to evaporate (or more volatile)
n) more flammable; burns with a clearer flame
o) higher boiling point range
p) darker in colour
q) more viscous
r) more difficult to evaporate (or less volatile)
s) less flammable; burns with a sootier flame

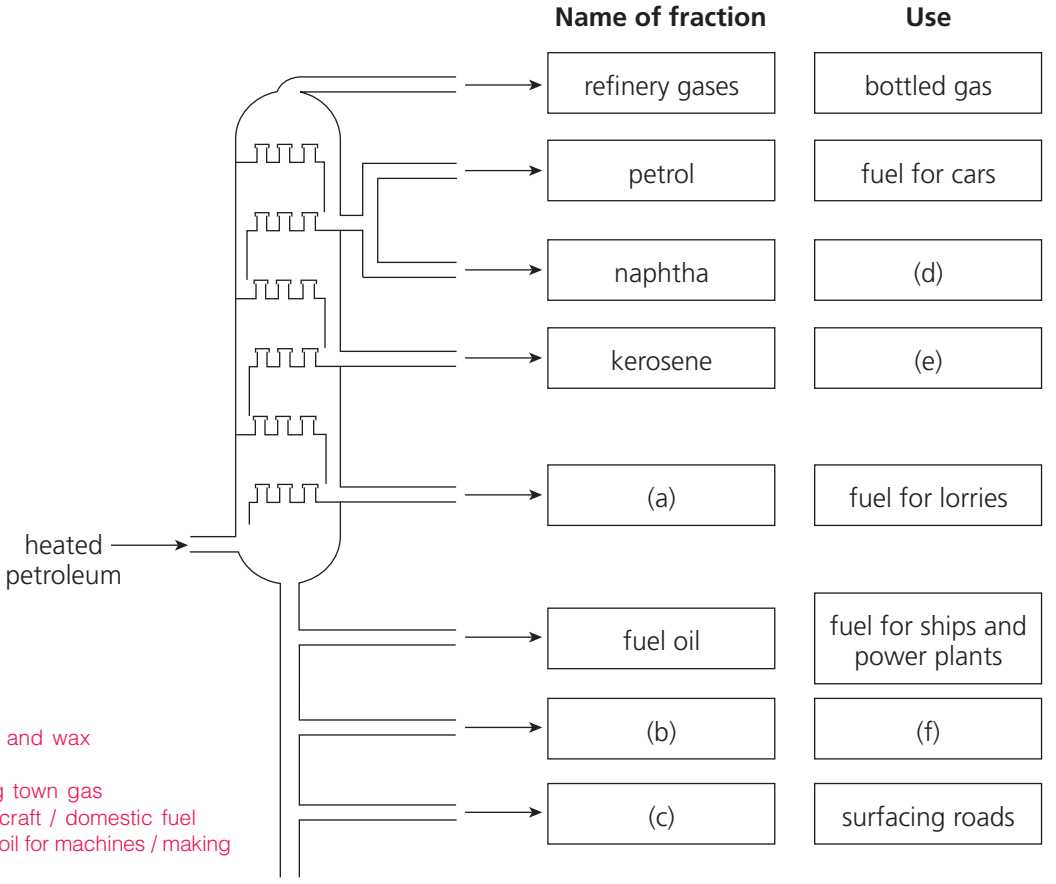
2 Complete the following concept map.

a) suspended particulates
b) carbon monoxide
c) unburnt hydrocarbons
d) oxides of nitrogen
e) sulphur dioxide
f) photochemical smog
g) acid rain



Unit Exercise

3 In an oil refinery, petroleum is separated into several fractions. Fill in the six spaces to complete the information on the diagram.



a) diesel oil
b) lubricating oil and wax
c) bitumen
d) manufacturing town gas
e) as fuel for aircraft / domestic fuel
f) as lubricating oil for machines / making candles

Part II Multiple choice questions

4 Which of the following petroleum fractions has the highest carbon content?

(A) Diesel
B Petrol
C Kerosene
D Naphtha

(HKCEE, Paper 2, 2008, 14)

5 Which of the following petroleum fractions is the most viscous?

A Diesel
(B) Fuel oil
C Kerosene
D Naphtha

6 Which of the following comparisons for fuel oil and naphtha is correct?

Refer to page T18 for explanation.

	Fuel oil	Naphtha
A	lighter in colour	darker in colour
B	more flammable	less flammable
C	very runny	quite thick
(D)	less volatile	more volatile

7 Which of the following combinations concerning the uses of petroleum fractions is correct?

	Fraction	Use
A	diesel oil	for surfacing roofs
(B)	kerosene	as a domestic fuel
C	naphtha	as a fuel for aircraft
D	refinery gas	for manufacturing town gas

8 Which of the following statements correctly describes an environmental problem caused by the burning of hydrocarbon fuels?

A The carbon dioxide is toxic and kills plants.
B The smoke produced reflects sunlight and leads to global warming.
C The water produced results in a damaging increase in rainfall.
(D) The carbon dioxide produced absorbs heat radiated from the Earth and leads to global warming.

(Edexcel Advanced Subsidiary GCE, Unit 1, Jun. 2009, 20)

9 Which of the following pairs is correctly matched?

Refer to page T18 for explanation.

	Pollutant	Effect
A	carbon monoxide	global warming
B	oxides of nitrogen	darkening of building walls
C	sulphur dioxide	photochemical smog
(D)	unburnt hydrocarbons	lung cancer

10 Which of the following statements concerning acid rain is INCORRECT?

A Acid rain refers to rain with pH less than 5.6.
B Acid rain can corrode iron window frames and marble buildings.
(C) One major air pollutant that causes the formation of acid rain is carbon dioxide.
D Acid rain will be formed when the gases discharged by power stations using fossil fuels enter the atmosphere.

(HKCEE, Paper 2, 2008, 1)

11 Which of the following statements about diesel is / are correct?

(1) Its boiling point range is lower than that of petrol.
(2) It is more volatile than fuel oil.
(3) It is commonly used to power heavy vehicles.

A (1) only
B (2) only
C (1) and (3) only
(D) (2) and (3) only

(1) The boiling point range of diesel is higher than that of petrol.

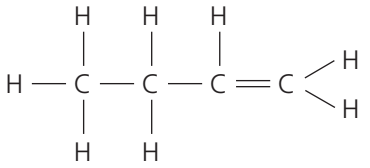
12 Which of the following substances can cause global warming?

(1) Carbon dioxide
(2) Methane
(3) Water vapour

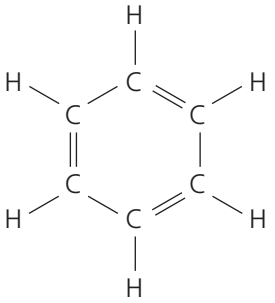
A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
(D) (1), (2) and (3)

13 Which of the following is / are aliphatic hydrocarbon(s)?

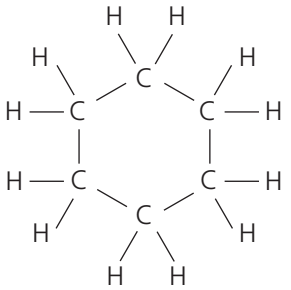
(1)



(2)



(3)



A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(2) Benzene is an aromatic hydrocarbon.

14 Which of the following statements concerning the measures to reduce air pollutants is / are correct?

(1) Scrubbers can be used to reduce carbon monoxide.
(2) Catalytic converters can be used to reduce nitrogen monoxide.
(3) Electrostatic precipitators can be used to reduce unburnt hydrocarbons.

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

(HKCEE, Paper 2, 2007, 26)

15 Which of the following environmental problems may be reduced by installing catalytic converters in petrol-driven cars?

(1) Acid rain
(2) Greenhouse effect
(3) Photochemical smog

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2010, 27)

16 Which of the following is / are measure(s) that can reduce the formation of acid rain?

(1) Installing scrubbers in power plants
(2) Using particulate removal devices in motor vehicles
(3) Installing electrostatic precipitators in power plants

A (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

16 (1) Sulphur dioxide can dissolve in rainwater to form acid rain. Installing scrubbers in power plants can reduce the emission of sulphur dioxide.
(2) Using particulate removal devices in motor vehicles can reduce the emission of particulates.
(3) Installing electrostatic precipitates in power plants can reduce the emission of particulates.

17 Any one of the following advantages:
- Coal energy is an affordable energy source because of coal's stable price compared to petroleum.
- Coal is easy to burn.
- Coal produces a lot of energy upon combustion.
- Coal is abundant.
- Coal is a reliable energy source.
- Coal is widely distributed all over the world.

Any two of the following disadvantages:
- Coal containing sulphur will burn to form sulphur dioxide which causes acid rain.
- Ash / smoke / soot / particulates formed.
- Sulphur burns to form sulphur dioxide which irritates respiratory system / affects functioning of lungs.

18 a) Hydrogen
It releases most energy. It produces water only when burnt.
b) Any one of the following:
- Whether it is readily available.
- How long it will last.
- Whether it is renewable.
- Whether it is easy to use.
- Whether it is easy to store.
- Whether it is toxic.
- Whether it is expensive.
- Whether it is volatile.
- Whether it is viscous.
- Whether it is a solid, liquid or gas.

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

17 Sulphur is a vital element in living organisms and a key industrial chemical. However, many covalent compounds of sulphur smell particularly bad.

Refer to the previous page.

Coal contains sulphur compounds. This has serious consequences for a society to decide whether to use coal as an energy source for generating electricity.

Suggest ONE advantage and TWO disadvantages of choosing coal, as compared to 'cleaner fossil fuels' such as methane, as an energy source.

(OCR Advanced Subsidiary GCE, Chem. B (Salters), F331, Jun. 2011, 3(c))

18 Look at the table. It gives information about the amount of energy released when some fuels burn. It also lists some of the products of burning.

Fuel	Energy released by one gram of fuel (kJ)	Products of burning
Biofuels (ethanol)	44.3	carbon dioxide and water
Hydrogen	143.0	water
Methane	55.6	carbon dioxide and water
Methanol	22.3	carbon dioxide and water
Petrol	48.3	carbon dioxide, water and other gases

Refer to the previous page.

a) Petrol can be used to power a car.

Choose ANOTHER fuel from the table that can be used to power a car. Explain your choice.

Use the information in the table.

b) Two factors to think about in choosing a fuel are the energy released and the products of burning.

Write down ONE other factor.

(OCR GCSE Gateway Science (Higher Tier), Chem. B, Unit 1, Jan. 2009, 3(a))

19 a) Petroleum fractions are used as raw materials to produce different chemicals in the petrochemical industry. These chemicals can be made into many useful products, such as synthetic polymers and detergents.

19 Petroleum is a mixture of many different compounds, each with a different boiling point.

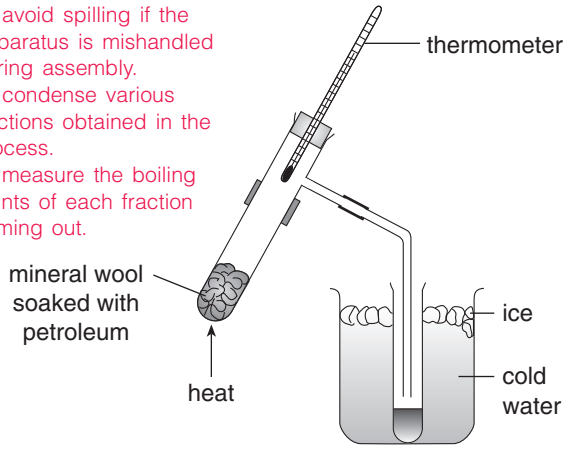
a) Petroleum is an important raw material. Why?

b) How can the compounds in petroleum be separated?

By fractional distillation

c) The following set-up can be used to conduct a simple separation in the laboratory.

c) i) To avoid spilling if the apparatus is mishandled during assembly.
ii) To condense various fractions obtained in the process.
iii) To measure the boiling points of each fraction coming out.



i) What is the purpose of the mineral wool?

ii) What is the purpose of the cold water?

iii) Why is the thermometer placed at the position shown?

d) Four fractions at different temperature ranges were collected in the above experiment:

Fraction	Temperature range
I	room temperature–100 °C
II	100 °C–150 °C
III	150 °C–200 °C
IV	200 °C–250 °C

i) Which fraction burns most easily?
Fraction I
ii) Which fraction is the most viscous?
Fraction IV
iii) Which fraction has the lightest colour?
Fraction I
iv) Which fraction is the most difficult to evaporate?
Fraction IV

20 The following substances are found in car exhaust:

carbon monoxide, carbon dioxide, oxides of nitrogen, hydrocarbons and particulates

Refer to page T19.

a) Explain why the following substances are present in car exhaust.

i) Carbon monoxide

ii) Oxides of nitrogen

b) State ONE harmful effect of oxides of nitrogen.

c) With the help of appropriate chemical equation(s), explain how a catalytic converter can reduce the emission of carbon monoxide and oxides of nitrogen.

d) A student wrote the following statement in an examination answer: 'Catalytic converters help reduce global warming.'

State whether you think that the statement is right or wrong. Explain your answer.

21 Crude oil is a mixture of many different compounds.

a) During industrial refining, crude oil is first separated into fractions.

Refer to page T19.

i) What is the name of the process used to obtain fractions from crude oil?

ii) Describe how the fractions are obtained.

b) Four of the fractions obtained from crude oil are:

bitumen, diesel, petrol, kerosene

i) Which fraction is the most viscous?
bitumen

ii) Which fraction is the most volatile?
petrol

iii) Which fraction is used in making roads?
bitumen

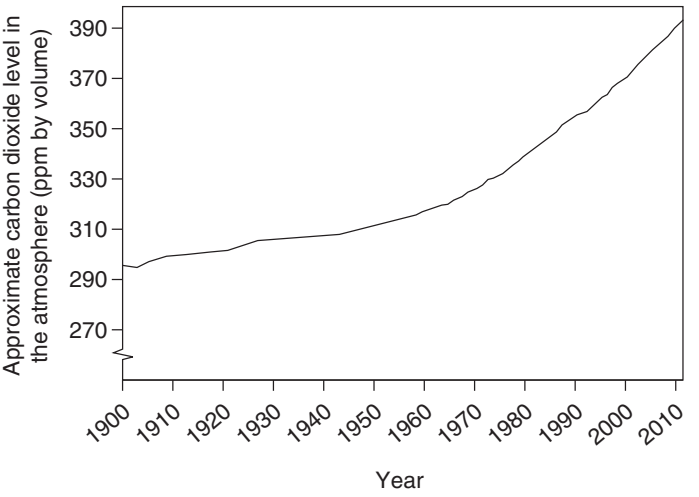
iv) Name TWO other fractions obtained from crude oil. refinery gases and fuel oil

(Edexcel IGCSE (Higher Tier), Chemistry, Paper 2H, Jun. 2010, 4(a)–(b))

22 Suggest THREE measures for reducing the emission of air pollutants upon using fossil fuels.

(HKDSE, Paper 1B, 2012, 10)

23 Carbon dioxide is a major greenhouse gas in the atmosphere. The following graph shows the approximate carbon dioxide level in the atmosphere from 1900 to 2012.



a) Suggest TWO reasons why there was a large increase in the concentration of carbon dioxide in the atmosphere in the past hundred years.

b) The presence of greenhouse gases in the atmosphere is important to life on Earth. However, too high a level of greenhouse gases in the atmosphere can cause global warming, which may lead to severe environmental consequences.

i) State the importance of greenhouse gases to life on Earth.

ii) State ONE severe environmental consequence associated with global warming.

iii) Suggest TWO possible ways to prevent further increase in the concentration of carbon dioxide in the atmosphere without sacrificing our present standard of living.

23 a) The number of motor vehicles increases rapidly. Burning a large amount of petrol / diesel produces carbon dioxide.
The rapid growth in population leads to deforestation, providing more land for housing. Thus, less carbon dioxide is removed from the atmosphere by photosynthesis.

b) i) The Earth absorbs the Sun's radiation and also radiates its own infrared radiation.
Greenhouse gases trap some of the infrared radiation which is radiated from the Earth, keeping the atmosphere warm for life to sustain on Earth.

ii) Increase in temperature of the atmosphere can cause melting of polar ice caps and glaciers at high altitude regions / flooding / change in rainfall pattern.

iii) Any two of the following:
- Reduce the use of fossil fuel by using alternative energy sources to generate electricity, e.g. wind energy.
- Use hydrogen as fuel in fuel cells for cars.
- Stop cutting down or burning forests.
- Plant more trees.